

Xueying Tang

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Education

- B.S. in Mathematics and Applied Mathematics, Peking University, Beijing, China, July 2011
 - Advisor: Dr. Jingping Yang
- M.S. in Statistics, University of Florida, Gainesville, FL, December 2013
 - Advisor: Dr. Nikolay Bliznyuk
 - Thesis: A Bayesian Spatio-Temporal Model for the Transmission of Infectious Diseases with Multiple Pathogens
- Ph.D. in Statistics, University of Florida, Gainesville, FL, August 2017
 - Advisors: Dr. Malay Ghosh and Dr. Nikolay Bliznyuk
 - Dissertation: Bayesian Analysis Under Shrinkage Priors

Employment

- Postdoctoral Research Scientist, Department of Statistics, Columbia Univeristy, July 2017 – June 2019
 - Advisors: Dr. Jingchen Liu and Dr. Zhiliang Ying
- Assistant Professor, Department of Mathematics, University of Arizona, August 2019 – July 2025
- Affiliate Member, Applied Mathematics Graduate Interdisciplinary Program, University of Arizona, September 2019 – present
- Member, Statistics & Data Science Graduate Interdisciplinary Program, University of Arizona, November 2019 – present
- Associate Professor, Department of Mathematics, University of Arizona, August 2025 – present

Publications

Refereed Journal Articles

1. Ghosh, P., **Tang, X.**, Ghosh, M., and Chakrabarti, A. (2016). Asymptotic Properties of Bayes Risk for a General Class of Shrinkage Priors in Multiple Hypothesis Testing under Sparsity, *Bayesian Analysis*, 11(3), 753–796.
2. **Tang, X.**, Li, K., Ghosh, M. (2017). Bayesian Multiple Testing Under Sparsity for Polynomial-Tailed Distributions. *Statistica Sinica*, 27, 1225–1242.
3. **Tang, X.**, Xu, X., Ghosh, M., and Ghosh, P. (2018). Bayesian Variable Selection and Estimation Based on Global-local Shrinkage Priors, *Sanhkya A*, 80(2), 215–246.
4. **Tang, X.**, Ghosh, M., Ha, N. S., and Sedransk, J. (2018). Modeling Random Effects Using Global-Local Shrinkage Priors in Small Area Estimation. *Journal of the American Statistical Association*, 113, 1476–1489.

5. **Tang, X.**, Chen, Y., Li, X., Liu, J., and Ying, Z. (2019) A Reinforcement Learning Approach to Personalized Learning Recommendation System. *British Journal of Mathematical and Statistical Psychology*, 72, 108–135.
6. **Tang, X.**, Yang, Y., Yu, H., Liao, Q., and Bliznyuk, N. (2019) A Spatio-Temporal Modeling Framework for Surveillance Data of Multiple Infectious Pathogens with Small Validation Sets. *Journal of the American Statistical Association*. 114, 1561–1573.
7. Merrill, H. R., **Tang, X.**, and Bliznyuk, N. (2019) Spatio-Temporal Additive Regression Model Selection for Urban Water Demand. *Stochastic Environmental Research and Risk Assessment. Stochastic Environmental Research and Risk Assessment*, 33, 1075–1087.
8. **Tang, X.**, Wang, Z., Liu, J., He, Q., and Ying, Z. (2020) Latent Feature Extraction for Process Data Using Multidimensional Scaling. *Psychometrika*, 85, 378–397.
9. Wang, Z., **Tang, X.**, and Liu, J. (2020) Statistical Analysis of Multi-Relational Network Recovery. *Frontiers in Applied Mathematics and Statistics*. doi: 10.3389/fams.2020.540225.
10. **Tang, X.**, Wang, Z., Liu, J., and Ying, Z. (2021) An Exploration of Process Data by Action Sequence Autoencoder. *British Journal of Mathematical and Statistical Psychology*. 74, 1–33.
11. **Tang, X.**, Wang, Z., Zhang, S., Liu, J., and Ying, Z. (2021) ProcData: An R Package for Process Data Analysis. *Psychometrika*. 86, 1058–1083.
12. Ghosh, T., Ghosh, M., Maples, J., and **Tang, X.** (2022) Multivariate Global-Local Priors for Small Area Estimation. *Stats*. 5(3). 673–688.
13. Lippitt, W., Sethuraman, S., and **Tang, X.** (2022) Stationarity and Inference in Multistate Promoter Models of Stochastic Gene Expression via Stick-Breaking Measures. *SIAM Journal on Applied Mathematics*. 82(6), 1953–1986.
14. Wang, Z., **Tang, X.**, Liu, J., and Ying, Z. (2023) Subtask Analysis of Process Data Through a Predictive Model. *British Journal of Mathematical and Statistical Psychology*. 76(1), 211–235.
15. **Tang, X.**, and Ghosh, M. (2023) Global-Local Priors for Spatial Small Area Estimation. *Calcutta Statistical Association Bulletin*. 75(2), 141–154.
16. **Tang, X.** (2024) A Latent Hidden Markov Model for Process Data. *Psychometrika*. 89(1), 205–240.
17. Zhang, S., **Tang, X.**, He, Q., Liu, J., and Ying, Z. (2024) External Correlates of Adult Digital Problem-Solving Process: An Empirical Analysis of PIAAC PSTRE Action Sequences. *Zeitschrift für Psychologie*. 232(2), 120–136.
18. **Tang, X.**, and Zhang, L. (2024) A Hierarchical Gamma Prior for Modeling Random Effects in Small Area Estimation. *Survey Methodology*. 50(2), 287–301.
19. Zhu, Z., and **Tang, X.** (2025) Modeling Sparsity with Super Heavy-Tailed Priors. *Electronic Journal of Statistics*. 19(1), 1570–1608.
20. **Tang, X.**, Liu, J., and Ying, Z. (2025) A Path Signature Perspective of Process Data Feature Extraction. *British Journal of Mathematical and Statistical Psychology*. 78(3), 939–964.
21. **Tang, X.**, and Fuquene-Patino, J. (2026) Shrinkage Priors for Functional and Structural Small Area Estimation with Applications to Family Planning Indicators. *Bayesian Analysis*. Forthcoming
22. Qiao, X., Liu, R., **Tang, X.**, Peterson, C. B., Jenq, R. R., and Zhang, L. (2026) Decoding Microbiome Dual Mediation: Introducing ZIMMA for Enhanced Zero-Inflated Data Analysis. *Annals of Applied Statistics*. Accepted.

Software

1. **Tang, X.**, Zhang, S., Wang, Z., Liu, J., and Ying, Z. (2020) R package ProcData.
2. **Tang, X.** (2024) R package proclhmm.

Book Review

1. **Tang, X.** (2025) Review of *Applied Machine Learning Using mlr3 in R* by Bernd Bischl, Raphael Sonabend, Lars Kotthoff, and Michel Lang. *The American Statistician*. 79 (2), 288–289.

Working Papers

1. Zhu, Z., **Tang, X.**, and Zhou, S. Bayesian Region Selection and Prediction in Poisson Regression with Spatially Dependent Global-Local Shrinkage Priors. *Major revision*.
2. Fuquene-Patino, J., Lewis, S., and **Tang, X.** A Bayesian Correlated Measurement Error Model for Small Area Estimation. *Major revision*.
3. **Tang, X.**, and Zhang, S. Hidden Markov Cognitive Diagnostic Models for Response Process Data. *In preparation*.
4. Yin, Z., Liu, C., Zhang, L., and **Tang, X.** Differential Omics Analysis based on the Zero-Inflated Negative Binomial Distribution. *In preparation*.
5. Zhao, Y. and **Tang, X.** Scalar-on-Process Regression: Modeling Time-Varying Action Effects in Response Process Data. *In preparation*.

Grants

1. Statistical Methods for Response Process Data, National Science Foundation, sole PI, \$161,680, 2023–2027

Teaching

- **Instructor** for MATH574M Statistical Machine Learning, Department of Mathematics, University of Arizona, Fall 2023
- **Instructor** for MATH566 Theory of Statistics, Department of Mathematics, University of Arizona, Spring 2021, Spring 2022, Spring 2023, Spring 2024
- **Instructor** for MATH571A Applied Linear Regression, Department of Mathematics, University of Arizona, Fall 2020, Fall 2024, Fall 2025, Fall 2026
- **Instructor** for MATH466 Theory of Statistics, Department of Mathematics, University of Arizona, Fall 2019 – Spring 2020, Fall 2021 – Spring 2023, Spring 2024, Fall 2025
- **Instructor** for STA3032 Engineering Statistics, Department of Statistics, University of Florida, Summer 2016
- **Teaching Assistant** for various undergraduate and graduate courses, Department of Statistics, University of Florida, Fall 2011 – Spring 2015
 - STA6166 Statistical Methods in Research 1, Spring 2015
 - STA6208 Experimental Design, Spring 2014, Spring 2015
 - STA6207 Applied Regression Analysis, Fall 2013, Fall 2014
 - STA2023 Introduction to Statistics 1, Summer 2013
 - STA3024 Introduction to Statistics 2, Summer 2012
 - STA4321/5325 Introduction to Probability Theory, Fall 2011, Spring 2012, Fall 2012, Spring 2013

Presentations

Invited Talks

1. International Chinese Statistical Association China Conference, Shenzhen, China, June 2026
2. Symposium on Data Science and Statistics, Madison, WI, April 2026
3. 19th International Joint Conference on Computational and Financial Econometrics and Computational and Methodological Statistics, Birkbeck, University of London, UK, December 2025
4. Biocomplexity Engineering Seminar, University of Florida, September 2025
5. 18th International Conference of the ERCIM WG on Computational and Methodological Statistics, King's College London, UK, December 2024
6. Joint Statistical Meetings, Portland, Oregon, August 2024
7. Arizona Data Science Day, University of Arizona, Tucson, AZ, April 2024
8. 16th International Conference of the ERCIM WG on Computational and Methodological Statistics, Berlin, Germany, December 2023
9. Psychometrics Workshop, Columbia University, New York, NY, September 2023
10. Seminar, Department of Statistics, University of Pittsburg, April 2023
11. Special Interest Group Seminar, Education Testing Service, February 2023
12. 15th International Conference of the ERCIM WG on Computational and Methodological Statistics, King's College London, UK, December 2022
13. Seminar, School of Mathematical and Statistical Sciences, Arizona State University, October 2022
14. Seminar, Department of Mathematical Sciences, University of Cincinnati, October 2022
15. International Chinese Statistical Association Applied Statistics Symposium, Gainesville, FL, June 2022
16. Seminar, Department of Statistics, University of Minnesota, April 2022
17. Arizona State University Machine Learning Day, April 2022
18. 14th International Conference of the ERCIM WG on Computational and Methodological Statistics, King's College London, UK, December 2021
19. 34th New England Statistics Symposium, Providence, RI, October 2021
20. Seminar, Department of Statistics, University of California Davis, June 2021
21. Seminar, Department of Biostatistics, The Ohio State University, April 2021
22. International Chinese Statistical Association Applied Statistics Symposium, virtual, December 2020
23. International Chinese Statistical Association Applied Statistics Symposium, Raleigh, NC, June 2019
24. TRIPODS Summer Research Conference, University of Arizona, May 2019
25. Seminar, Department of Statistics, George Washington University, March 2019
26. Seminar, Department of Mathematics, University of Arizona, March 2019
27. Seminar, Department of Statistics, University of California, Irvine, February 2019
28. Seminar, Department of Statistics, Iowa State University, January 2019
29. Seminar, School of Mathematical and Statistical Sciences, Arizona State University, January 2019
30. Seminar, Department of Statistics and Applied Probability, University of California, Santa Barbara, January 2019
31. Colloquium, Department of Mathematics, University of Maryland, College Park, January 2019
32. ENAR Spring Meeting, Washington D.C., March 2017
33. Seminar, Department of Mathematical Sciences, University of Arkansas, December 2016

Contributed Talks

1. Joint Statistical Meetings, Nashville, Tennessee, August 2025
2. Joint Statistical Meetings, Toronto, Ontario, Canada, August 2023
3. International Meetings of Psychometric Society, College Park, MD, July 2023
4. International Meetings of Psychometric Society, virtual, July 2020
5. Joint Statistical Meetings, virtual conference, August 2020
6. International Meetings of Psychometric Society, Santiago, Chile, July 2019
7. Joint Statistical Meetings, Vancouver, BC, Canada, August 2018
8. International Meetings of Psychometric Society, New York, NY, July 2018
9. Joint Statistical Meetings, Chicago, IL, August 2016
10. Fifth International Conference on Infectious Disease Dynamics, Clearwater, FL, December 2015
11. Student Seminar Series, Department of Statistics, University of Florida, Gainesville, FL, March 2015
12. ASA Florida Chapter Meeting, Tampa, FL, February 2015
13. ASA Florida Chapter Meeting, Gainesville, FL, February 2014

Contributed Posters

1. International Workshop on Objective Bayes Methodology, Austin, TX, December, 2017
2. ASA Florida Chapter Meeting, Tallahassee, FL, February 2016
3. SRCOS Summer Research Conference, Wilmington, NC, June, 2015
4. ENAR Spring Meeting, Miami, FL, March 2015
5. EPI Research Day, Emerging Pathogens Institute, University of Florida, Gainesville, FL, February 2015

Short Courses

1. Instructor, Process Data Workshop, virtual, Summer 2020
2. Instructor, Statistical Learning Methods for Process Data, International Meetings of Psychometric Society, virtual, Summer 2021
3. Instructor, Statistical Learning of Process Data: Methods, Software, and Applications, NCME Annual Meeting, virtual, Summer 2021

Honors and Awards

- GlaxoSmithKline Scholar Award, Department of Statistics, University of Florida, Fall 2011
 - Awarded to the best incoming Ph.D. student
- William Mendenhall Award, Department of Statistics, University of Florida, Fall 2012
 - Awarded to the best first-year Ph.D. student
- Bagui Award, Florida Chapter of the American Statistical Association Annual Meeting, February 2015
 - Awarded to the best student papers and presentations
- Travel Award, Office of Research, University of Florida, Summer 2016

- Travel Award, College of Liberal Arts and Sciences, University of Florida, Summer 2016
- Travel Award, SRCOS Summer Research Conference, June 2015
- Statistics Faculty Award, Department of Statistics, University of Florida, Fall 2016
 - Awarded to the best graduating student
- Travel Award, International Workshop on Objective Bayes Methodology, December 2017
- Outstanding Reviewer, American Educational Research Association and Journal of Educational and Behavioral Statistics, March 2021
- Elected to Arizona Alpha Chapter of Mu Sigma Rho, the National Honor Society in Statistics, 2024
- Best Reviewer Award, the Psychometric Society, 2024

Service

Departmental Committees

- Co-chair, Mathematics Colloquium Committee, Department of Mathematics, University of Arizona, July 2020 – June 2021
- Member, Qualifying Exam Committee, Statistics & Data Science GIDP, University of Arizona, August 2020 – present
- Member, Admission Committee, Statistics & Data Science GIDP, University of Arizona. Spring 2022, Spring 2023
- Member, Statistics & Data Science Major Working Group, University of Arizona. Spring 2022 – present
- Member, Graduate Committee, Department of Mathematics, University of Arizona. Fall 2023 – Fall 2024
- Interim Chair, Qualifying Exam Committee, Statistics & Data Science GIDP, University of Arizona, Spring 2024
- Chair, Qualifying Exam Committee, Statistics & Data Science GIDP, University of Arizona, Fall 2024 – present
- Member, Executive Committee, Statistics & Data Science GIDP, University of Arizona, Spring 2024 – present

Conference Organization

- Session chair, International Meeting of the Psychometric Society, virtual, July 2020
- Co-organizer, Workshop on Statistical Learning for Process Data, virtual, July 2020
- Session organizer and chair, International Chinese Statistical Association Applied Statistics Symposium, virtual, December 2020
- Session organizer and chair, International Chinese Statistical Association Applied Statistics Symposium, Gainesville, FL, June 2022
- Session chair, International Meeting of the Psychometric Society, University of Maryland, College Park, MD, July 2023
- Session organizer and chair, 19th International Joint Conference on Computational and Financial Econometrics and Computational and Methodological Statistics, Birkbeck, University of London, UK, December 2025

Professional Service

- Editorial Board, Journal of Educational and Behavioral Statistics. July 2022 – present
- Referee for Annals of Applied Statistics (2), Bayesian Analysis (2), Behavior Research Methods (1), Biometrics (3), British Journal of Mathematical and Statistical Psychology (4), Computational Statistics and Data Analysis (1), Electronic Journal of Statistics (2), Frontiers in Psychology (1), Journal of Business and Economic Statistics (1), Journal of Computational and Graphical Statistics (1), Journal of Data Science (2), Journal of Educational and Behavioral Statistics (10), Journal of Educational Measurement (1), Journal of Intelligence (2), Journal of Multivariate Analysis (2), Journal of the American Statistical Association (2), Journal of Royal Statistical Society Series A (1), Journal of Royal Statistical Society Series C (1), Journal of Statistical Computation and Simulation (1), Journal of Survey Statistics and Methodology (1), Multivariate Behavioral Research (1), PLOS ONE (3), Psychometrika (12), Statistics (1), Statistical Methods & Applications (1), Statistics and Probability Letters (1), Survey Methodology (1), The American Statistician (2), Zeitschrift für Psychologie (1)
- External Reviewer, Hong Kong Research Grants Council, February 2023, February 2024
- Reviewer, National Science Foundation, November 2023, January 2025
- Reviewer, AERA Annual Meeting, September 2024
- Reviewer, NCME Training Session, October 2024

Outreach

- Tutorial Session Instructor and Judge, Arizona ASA DataFest, March 2022
- Tutorial Session Instructor and Judge, Arizona ASA DataFest, March 2023
- Tutorial Session Instructor and Mentor, Arizona ASA DataFest, April 2024

Student Advisory

Ph.D. Dissertation

1. Ah Young Kim (Ph.D. Summer 2023). *Dynamic Structure Estimation of Time-Varying Networks*. Statistics & Data Science GIDP, University of Arizona. (co-advised with Hao Helen Zhang)
2. Daniella Van Boxell (Ph.D. Spring 2024). *Bayesian Additive Regression Networks (BARN): A Machine Learning Algorithm and Scientific Software Library*. Applied Mathematics GIDP, University of Arizona. (co-advised with Cristian Román-Palacios in School of Information)
3. Zihan Zhu (Ph.D. Summer 2024). Statistics & Data Science GIDP, University of Arizona

Master Thesis

1. Drew Baldwin (M.S. Winter 2020). *Spatiotemporal Bayesian Model of PM levels in Korea*. Statistics & Data Science GIDP, University of Arizona.

Undergraduate Honor Thesis

1. Josh Colten (B.S. Spring 2023). *Can You Beat the Odds? A Reinforcement Learning Approach to Optimal Policy Search in Blackjack*. Department of Mathematics, University of Arizona.

Other (Dissertation/Comprehensive Exam/Thesis Committee Member)

1. Cameron LeBlanc (M.S. Summer 2021). Statistics & Data Science GIDP, University of Arizona.
2. Kyung Mi Chung (Ph.D. Summer 2022). Statistics & Data Science GIDP, University of Arizona.
3. Jocelyn Rios (Ph.D. Fall 2022). Department of Mathematics, University of Arizona.

4. Zhiming Zhong (Ph.D. Spring 2024). Department of Systems and Industrial Engineering, University of Arizona.
5. Alonso Granados Baca (Ph.D. candidate). Department of Computer Science, University of Arizona.
6. Hao Qin (Ph.D. candidate). Statistics & Data Science GIDP, University of Arizona.
7. Apurba Kumar Saha (Ph.D. Fall 2024). Department of Systems and Industrial Engineering, University of Arizona.
8. Elliot Owen Kennedy (M.S. Summer 2024). Statistics & Data Science GIDP, University of Arizona.
9. Taryn Laird (M.S. Summer 2024). Statistics & Data Science GIDP, University of Arizona.
10. Sunday Oluwadamilola Usman (Ph.D. candidate). Department of Systems and Industrial Engineering, University of Arizona.
11. Kapilan Balagobalan (Ph.D. candidate). Department of Computer Science, University of Arizona.
12. John Park (Ph.D Fall 2025). Department of Mathematics, University of Arizona.
13. Yueqing (Miranda) Li (Ph.D. candidate) Department of Management Information Systems, University of Arizona.
14. Salma Khatoon (M.S. Summer 2026). Statistics & Data Science GIDP, University of Arizona.
15. Cody Melcher (Ph.D. Summer 2026). Statistics & Data Science GIDP, University of Arizona.

Last updated: August 19, 2026